

Privacy & Data Retention Policy

Enhancing Security in Messaging Applications

Spinnaker Analytics

29 June 2026

Table of Contents

1. Introduction
2. Privacy by Design Principles
3. Data Classification
4. Data Collection & Processing
5. Data Retention Policy
6. Secure Data Storage & Disposal
7. User Privacy Rights
8. Privacy Compliance & Security Controls
9. Privacy Validation & Auditing
10. Conclusion

1. Introduction

1.1 Purpose

- This document defines the privacy and data retention controls implemented within the messaging application.
- The objective is to protect user privacy by minimizing data collection, restricting retention periods, and ensuring secure handling and disposal of information throughout its lifecycle.

1.2 Objectives

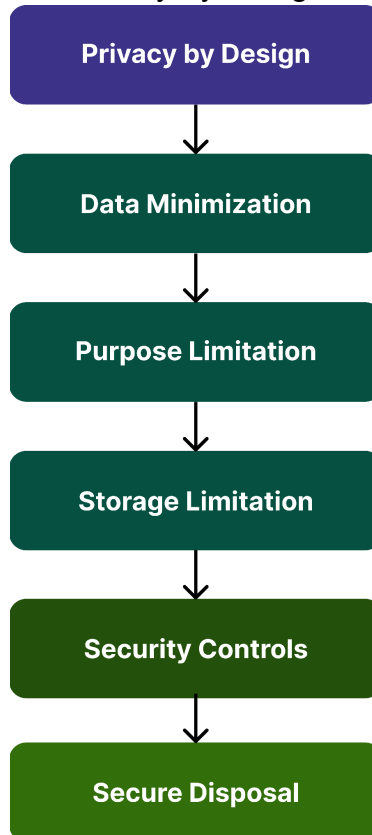
- Apply Privacy-by-Design principles
- Minimize personal data collection
- Define retention schedules
- Protect stored metadata
- Support secure deletion
- Ensure regulatory compliance
- Protect user privacy throughout the data lifecycle

2. Privacy by Design Principles

The messaging application incorporates privacy into every stage of the system lifecycle.

2.1 Privacy by Design

Figure 27 – Privacy-by-Design Framework



2.2 Core Principles

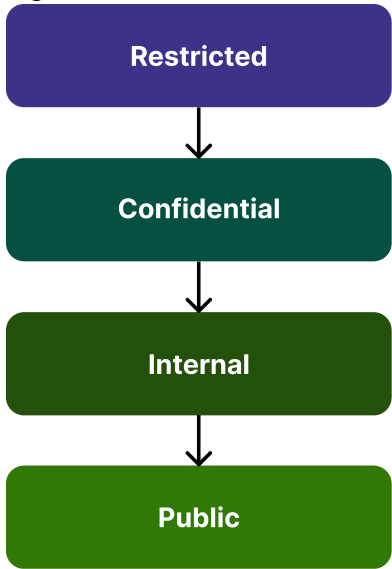
Principle	Implementation
Data Minimization	Only essential metadata collected
End-to-End Encryption	Message contents inaccessible to server
Purpose Limitation	Data used only for messaging functions
Storage Limitation	Automatic retention schedules
Integrity & Confidentiality	Encryption in transit and at rest
Accountability	Auditing and logging

3. Data Classification

3.1 Information Categories

Data Type	Classification	Protection
Message Content	Confidential	End-to-End Encryption
Voice Calls	Confidential	SRTP
Files	Confidential	AES-256-GCM
User Profile	Internal	Encrypted Database
Authentication Data	Restricted	Argon2id + MFA
System Logs	Internal	Access Controlled

Figure 28 – Data Classification Model



4. Data Collection & Processing

4.1 Data collected

Information	Purpose
Username	Account identification
Phone Number	User registration
Public Keys	End-to-end encryption
Device Identifier	Session management
Push Notification Token	Message notifications

4.2 Data Not Collected

- Message contents
- Voice recordings
- File contents
- Encryption keys
- Passwords in plaintext

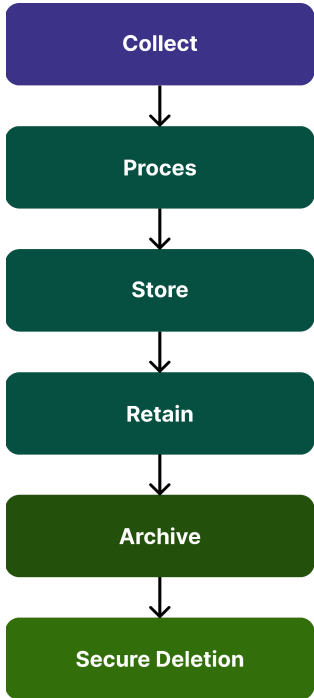
5. Data Retention Policy

5.1 Retention Schedule

Data Type	Retention Period
Account Information	Until account deletion
Metadata	30 Days
Authentication Logs	90 Days
Security Logs	180 Days
Incident Reports	1 Year
Encrypted Backups	30 Days

5.2 Data Lifecycle

Figure 29 – Data Lifecycle



6. Secure Data Storage & Disposal

6.1 Storage Controls

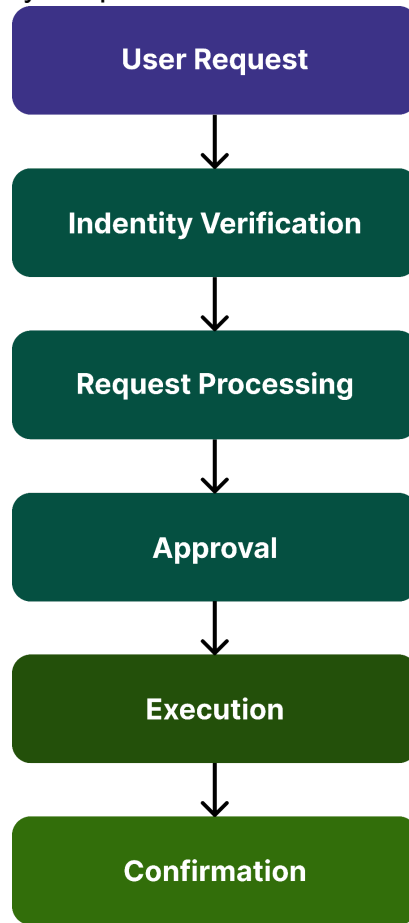
- AES-256 encrypted databases
- SQLCipher for local storage
- Secure backups
- Role-based access control
- Key management through Secure Enclave / Android Keystore

6.2 Secure Deletion

Data Type	Deletion Method
Messages	Cryptographic Erasure
Files	Secure Overwrite
Metadata	Database Purge
Backups	Automated Expiry

7. User Privacy Rights

- Users may:
 - View stored account information
 - Request correction of personal information
 - Export account data
 - Delete their account
 - Withdraw consent where applicable
- Figure 30 – User Privacy Request Workflow



8. Privacy Compliance & Security Controls

8.1 Security Controls

Control	Purpose
TLS 1.3	Secure transmission
AES-256-GCM	Data confidentiality
Signal Protocol	End-to-end encryption
RBAC	Access restriction
MFA	Administrative protection
Audit Logging	Accountability

8.2 Standards Alignment

Standard	Purpose
GDPR	Data protection principles
ISO/IEC 27001	Information Security Management
NIST Privacy Framework	Privacy risk management
OWASP MASVS	Mobile application security

9. Privacy Validation & Auditing

9.1 Validation Tools

Tool	Purpose
MobSF	Mobile privacy assessment
Wireshark	Verify encrypted traffic
Wazuh	Log monitoring
OpenSSL	Certificate verification

9.2 Test Cases

Test ID	Scenario	Expected Result
PRIV-01	Message inspection	No plaintext visible
PRIV-02	Metadata review	Minimal metadata retained
PRIV-03	Account deletion	Data removed
PRIV-04	Backup expiry	Automatic deletion
PRIV-05	Traffic capture	Encrypted communications

10. Conclusion

- This Privacy & Data Retention Policy establishes a privacy-first framework for the messaging application.
- By applying Privacy-by-Design principles, limiting data collection, enforcing defined retention schedules, protecting stored information with strong cryptography, and supporting secure deletion, the application safeguards user information while maintaining compliance with recognized security and privacy practices.